Experiment Results:
Bilateral Asymmetric Information Bargaining (Baseline)

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Research Question

Under bilateral asymmetric information, do LLM-simulated bargaining agents reach efficient deals, and does the agreed price reflect each party's private information as standard theory predicts?

In this game, a buyer (simulated human, `gpt-5-nano`) and an AI seller (`gpt-5`) negotiate over an item via alternating offers. Each party knows only their own private value: the buyer knows their valuation $v \in \{60, 70, 80, 90\}$ but not the seller's cost $c \in \{20, 30, 40, 50\}$, and vice versa. Both know the opponent's value is uniform over the four-element set. The zone of possible agreement (ZOPA) is always positive ($v - c \in [10, 70]$, mean = \$43.50), so efficient deals are always feasible.

The baseline prediction from standard Bayesian bargaining theory is: (1) deal rates near 100% given large ZOPAs; (2) deal prices that increase in buyer value (buyers with higher surplus can concede more) and in seller cost (sellers with higher cost demand more); (3) deal prices close to the Nash bargaining solution (midpoint of ZOPA). No hypothesis memo exists for this game — this is a baseline characterization of LLM behavior under bilateral asymmetric information.

Experimental Design

Game structure. Single condition (no between-subjects manipulation). The buyer offers on odd rounds (1, 3, 5); the AI offers on even rounds (2, 4). Maximum 5 rounds; no deal yields \$0 for both. Parameters are randomized per session: v and c drawn independently and uniformly from their respective sets.

Variable	Values	Type
`buyer_value` (v)	\$60, \$70, \$80, \$90	Randomized per session
`seller_cost` (c)	\$20, \$30, \$40, \$50	Randomized per session
`zopa`	$v - c$	Derived
`fair_price`	$(v + c)/2$	Derived (midpoint)

Table 1: Game variables. All four buyer values and all four seller costs are possible in each session.

Sample. $N = 20$ sessions run with `interview simulate bargain_asymmetric -n 20`. Design: random assignment (not factorial), yielding unequal cell counts across the 16 parameter combinations.

Data Overview

Sample. $N = 20$ sessions total. Condition distribution: `buyer_value` ranged from 4 sessions at \$60 to 9 at \$70 (unbalanced due to random draw). `seller_cost` ranged from 2 sessions at \$50 to 8 at \$20. ZOPA range: \$20–\$70, mean = \$43.50.

Checker results. 19/20 nominal pass (95%). One session (`sim13`) failed the checker; inspection shows a checker false positive on arithmetic (human earns \$10 = \$60 − \$50, which is correct). No substantive game failures were detected.

Metric	Value	SD
Deal rate	100% ($N = 20$)	—
Mean deal price	\$54.60	\$5.96
Mean fair price (midpoint)	\$51.75	\$7.29
Price deviation from fair	+\$2.85	—
Mean human earnings	\$18.40	\$10.64
Mean AI earnings	\$25.10	\$13.40
Buyer surplus share	0.450	0.260
Mean rounds to deal	4.30	1.23
First offer (buyer)	\$50.00	\$5.00

Table 2: Summary statistics across all 20 sessions.

Results

Primary Outcome: Deal Rate

All 20 sessions reached a deal (100%). This confirms that large ZOPAs (mean = \$43.50) make efficient agreement readily achievable even under bilateral asymmetric information, consistent with the theoretical baseline.

Deal Price and Surplus Split (Figure 1)

The mean deal price was \$54.60 (SD = \$5.96), versus a mean fair-price midpoint of \$51.75 (SD = \$7.29). The mean deviation of +\$2.85 (AI-favorable) was not statistically significant ($t(19) = 1.21$, $p = 0.243$). Deal prices exceeded the fair-price midpoint in 14 of 20 sessions, consistent with a mild AI (seller) advantage but not significant at conventional levels.

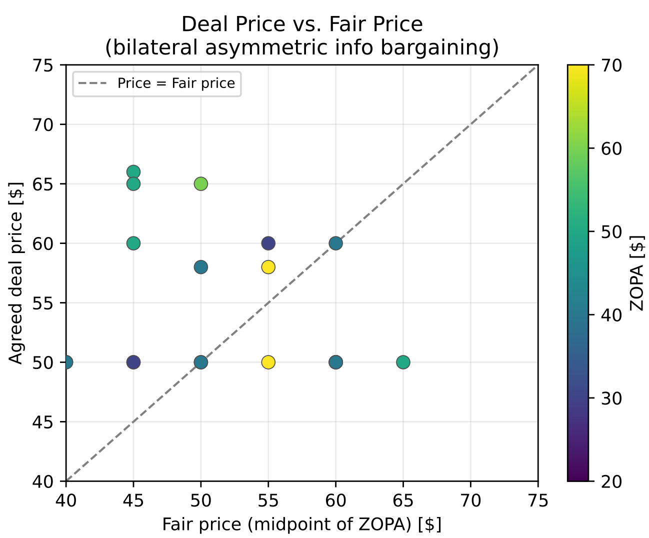

Figure 1: Deal price vs. fair-price midpoint ($(v + c)/2$) for each session. Points above the dashed diagonal indicate AI-favorable outcomes (price above midpoint). Color encodes ZOPA size. Most points cluster slightly above the diagonal.

Surplus Split by Condition (Figure 2)

The AI (seller) earned a mean of \$25.10 vs. \$18.40 for the human (buyer), capturing approximately 55% of the total surplus on average. A paired t-test found the AI advantage to be marginally significant (one-tailed $p = 0.079$). The buyer surplus share was 0.45 (SD = 0.26), below the equal-split benchmark of 0.50, though the distribution is wide.

By seller cost: mean deal prices were \$58.00 (sc = \$20), \$51.33 (sc = \$30), \$55.00 (sc = \$40), and \$50.01 (sc = \$50). The negative relationship (higher seller cost → lower deal price) is counter to theory and is explained by the buyer's anchoring behavior described below.

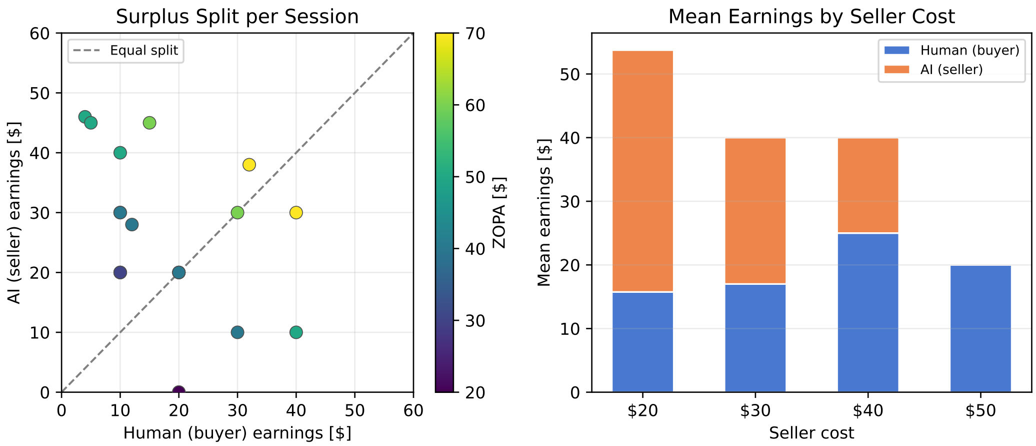

Figure 2: *Left:* Buyer vs. AI earnings per session; dashed line = equal split. *Right:* Mean earnings by seller cost. Error bars omitted due to small cell sizes.

Parameter Effects and Bargaining Dynamics (Figure 3)

The most striking finding is that deal price is nearly unresponsive to the game parameters (OLS regression: $R^2 = 0.14$; buyer-value coefficient = 0.054, seller-cost coefficient = -0.223). Standard theory predicts that deal prices should increase in seller cost (minimum acceptable price) and increase in buyer value (maximum willingness to pay), producing R^2 near 1.0. Neither prediction holds.

The source of this finding is clear from Figure 3: the simulated buyer (`gpt-5-nano`) consistently anchors at \$50.00 as its opening offer. Specifically, 11 of 16 opening offers were exactly \$50.00 and the mean opening offer was \$50.00 (SD = \$5.00), *regardless of the buyer's true value* ($v \in \{60, 70, 80, 90\}$). This means a buyer with value $v = 90$ opens at the same price as a buyer with value $v = 60$ — a massive failure to condition on private information.

Consequence: most negotiations were highly positional. The mode was 5 rounds (14/20 sessions), as the AI repeatedly countered above \$50 and the buyer refused to update. The AI eventually capitulated in Round 5 to avoid the disagreement payoff of \$0. In two sessions with seller cost = \$50, the AI accepted at exactly \$50.00 — earning \$0 — rather than taking the outside option of \$0. This constitutes a clear AI decision-quality failure.

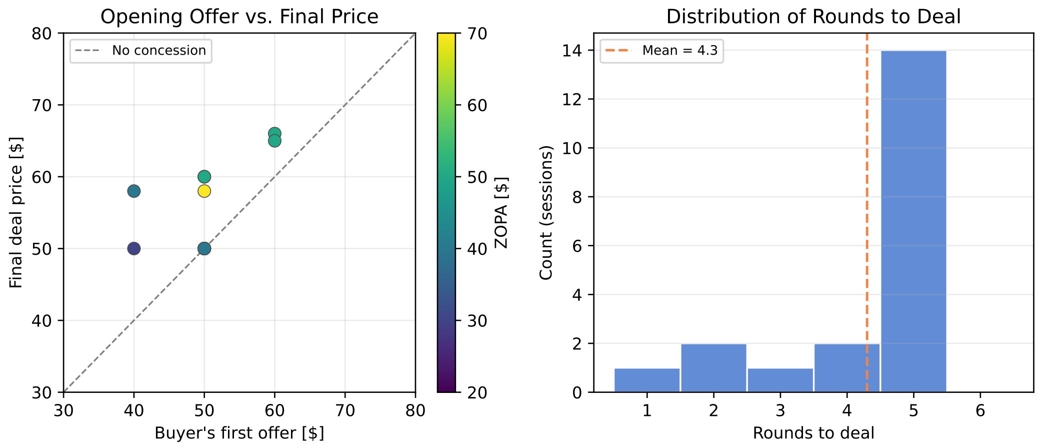

Figure 3: *Left:* Buyer's opening offer vs. final deal price. Most buyers open at \$50 regardless of their value; final prices vary due to AI concession behavior. *Right:* Distribution of rounds to deal; mode = 5, meaning most games go to the final round.

Evaluation

Did We Learn Something Interesting?

Partially — we found a clear behavioral anomaly but the design conflates two sources.

The interesting finding is: **the gpt-5-nano buyer anchors rigidly at \$50, failing to condition offers on its private value.** This is not rational information-hiding (a buyer *could* strategically low-ball regardless of value); it appears to be a genuine failure of value-based reasoning by the smaller model. The resulting deal prices are nearly constant (\$50–\$66 range) despite a 4× variation in ZOPA (\$20–\$70).

However, what we learned is unclear for two reasons:

1. **Model conflation.** The buyer uses `gpt-5-nano` and the AI seller uses `gpt-5`. The behavioral asymmetry we observe (buyer rigid, AI concedes) could reflect: (a) the gpt-5-nano model's limited strategic capacity, (b) a game design issue in `sim_human.md` that anchors the buyer at \$50, or (c) rational buyer strategy. We cannot distinguish without isolating each factor.

2. **No baseline condition.** We do not know what the same game looks like with symmetric information or with `gpt-5` on both sides, so we cannot attribute the behavior to the information structure.

Score against the triviality dimensions:

- *Surprising?* Yes — the near-zero price sensitivity ($R^2 = 0.14$) was unexpected.
- *Fills a gap?* Partially — documents LLM bargaining behavior under bilateral asymmetry.
- *Mechanism clear?* No — model vs. prompt vs. rationality confounded.
- *Credible evidence?* Low — $N = 20$ is small, randomization was unbalanced, no control condition.

Limitations

- **Unbalanced design.** Random draw yielded 9 sessions at $v = 70$ and only 2 at $c = 50$. A factorial design (16 cells, 2–3 reps each) would better identify parameter effects.
- **Buyer model.** The sim_human uses `gpt-5-nano`, a smaller model likely less capable of strategic reasoning. The \$50 anchor may reflect this model's limitations rather than the information structure.
- **AI decision quality.** The AI seller accepted a zero-profit deal in 2 sessions ($c = 50$, $p =$ \$50.00). This violates rational play. A better AI player prompt should include "never accept a price that leaves you with zero or negative profit unless all rounds are exhausted."
- **No baseline.** Without a symmetric-information condition, we cannot attribute deal price patterns to the asymmetric information structure.

Next Experiment

Archetype: REFINE. The game design has two issues that prevent clean inference: (1) buyer model (`gpt-5-nano`) is too weak to reason about private values, and (2) no between-subjects manipulation. The next experiment should run the same bargaining game with `gpt-5` as both the buyer and seller, and compare symmetric vs. asymmetric information conditions to test whether information structure (not model capability) drives the price-insensitivity result.

Proposed next experiment: "Does bilateral information asymmetry reduce surplus efficiency in LLM bargaining, relative to a symmetric-information baseline?"

- **Condition A (Asymmetric):** Same as current game — neither side knows the other's value.
- **Condition B (Symmetric):** Both sides know both v and c (full information; both agents calculate the Nash midpoint).
- **Buyer model:** `gpt-5` for both sides (use `--model gpt-5` in the sim_human prompt).
- **Design:** Factorial $2 \times 4 \times 4$ (info $\times$ v $\times$ c), at least 2 reps per cell $\Rightarrow N \approx 64$.
- **Primary test:** Does deal price respond to v and c under symmetric info but not asymmetric info?
- **Expected finding:** Under symmetric info, R^2 near 1.0 (prices $\approx$ midpoint always); under asymmetric info, lower R^2 but similar deal rate. If gpt-5 buyer also anchors at \$50, the anomaly is confirmed as a model artifact; if not, it was a gpt-5-nano limitation.

Run: `/hypothesize iterate bargain_asymmetric` to formalize this hypothesis, or `interview simulate bargain_asymmetric_gpt5 -n 64 --design factorial` after creating the revised game files.

Prior Experiment Context (Machine-Readable)

prior_hypothesis: Under bilateral asymmetric information, LLM bargaining agents
  reach efficient deals and deal prices reflect private values per Bayesian theory.
verdict: PARTIALLY
diagnosis: IMPLEMENTATION
key_finding: 100% deal rate but deal prices show near-zero sensitivity to private
  values (R2=0.14); gpt-5-nano buyer anchors rigidly at $50 regardless of true value.
key_statistic: R2=0.14 for deal_price ~ buyer_value + seller_cost; mean first
  offer = $50 (SD=$5) for buyer_value in {60,70,80,90}
dag_variables: X=information_structure, M=offer_anchoring, Y=deal_price, Z=ZOPA
testable_implications_results: deal_rate_near_100=CONFIRMED,
  price_increases_in_buyer_value=REFUTED, price_increases_in_seller_cost=REFUTED,
  deal_near_fair_price=PARTIALLY_CONFIRMED(p=0.24)
next_archetype: REFINE
proposed_changes: Use gpt-5 for both buyer and seller; add symmetric-information
  baseline condition; use factorial design with 2+ reps per cell (N~64).
next_hypothesis_sketch: Under bilateral asymmetric information, LLM bargainers
  (gpt-5) show lower price sensitivity to private values than under symmetric
  information, revealing the causal role of information structure vs. model capacity.